

INTRODUCTION

Time series forecasting predicts future values from historical data and is used in energy, traffic, and weather systems.

Challenges:

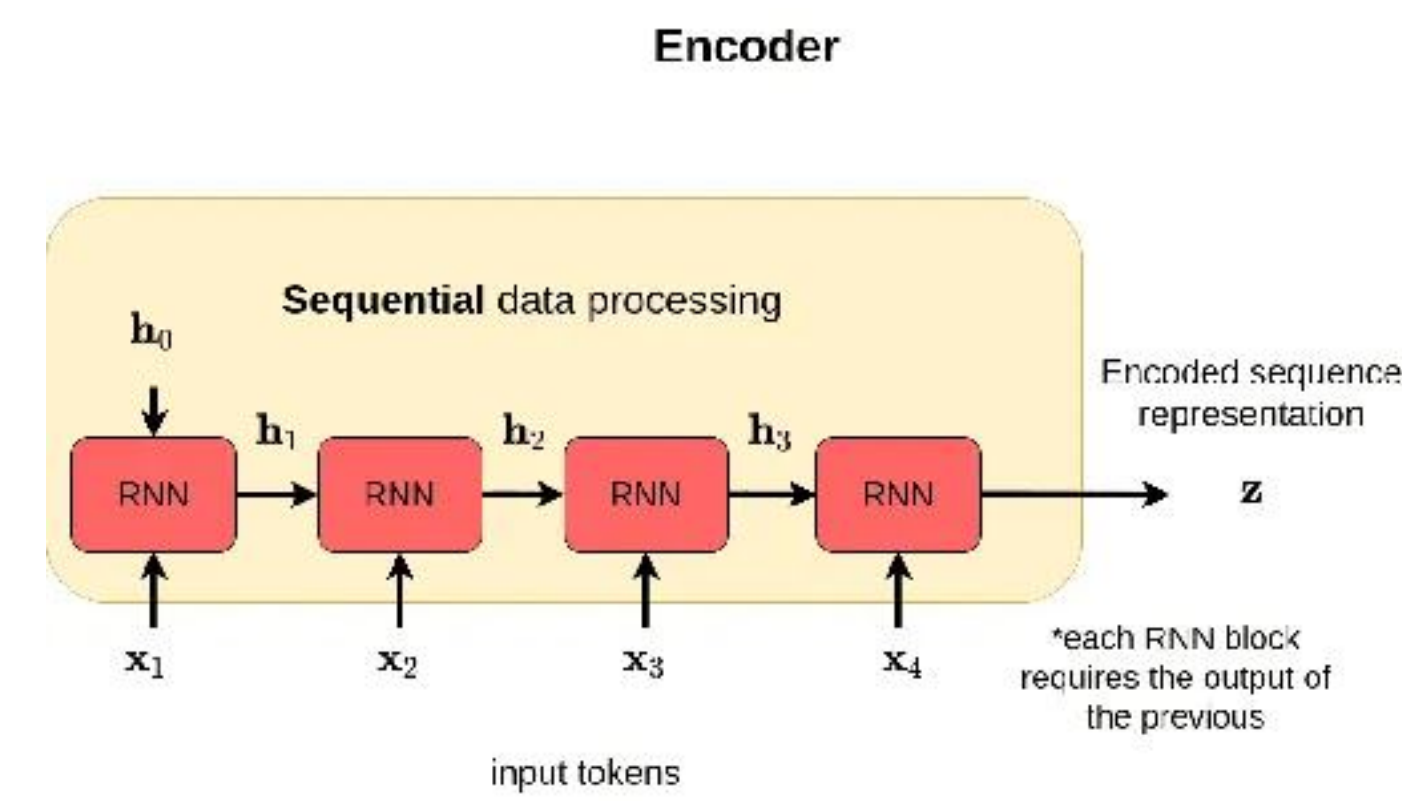
- Hard to capture long-range dependencies in long sequences
- RNN/LSTM models are sequential → slow and hard to scale

Transformers:

- Model global dependencies with attention
- Use point-wise tokens (individual timesteps)

Limitation:

- Single timesteps lack semantic meaning
- Attention over long sequences is computationally expensive ($O(n^2)$)



GOAL

Design a **patch-based Transformer (PatchTST)** that:

- Uses subsequence (patch) representations instead of individual timesteps
- Enables efficient attention over longer sequences
- Improves long-term forecasting performance over existing models

SETUP

Datasets: ETTh1, ETTh2, ETTm1, ETTm2, Weather, ILI

Task: Multivariate long-term forecasting

Input / Output:

- Look-back window: $L = 336$ ($L = 104$ for ILI)
- Forecast horizons: $T \in \{96, 192, 336, 720\}$ ($T \in \{24, 36, 48, 60\}$ for ILI)

Preprocessing:

- Global normalization (train-set mean/std)
- RevIN (per-instance normalization with learnable affine)

Metrics:

- MSE, MAE (evaluated in globally-normalized scale)

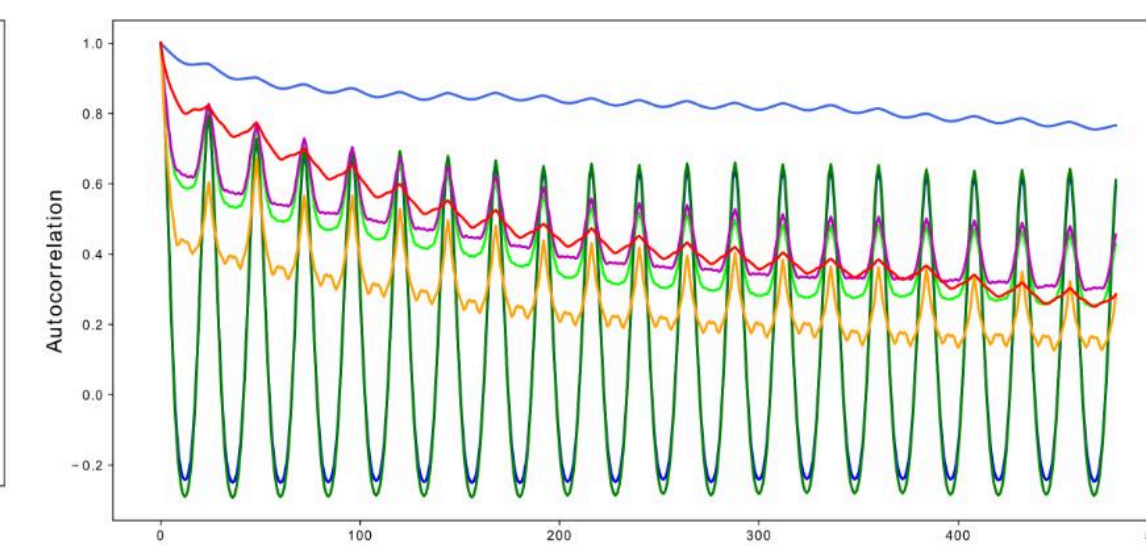
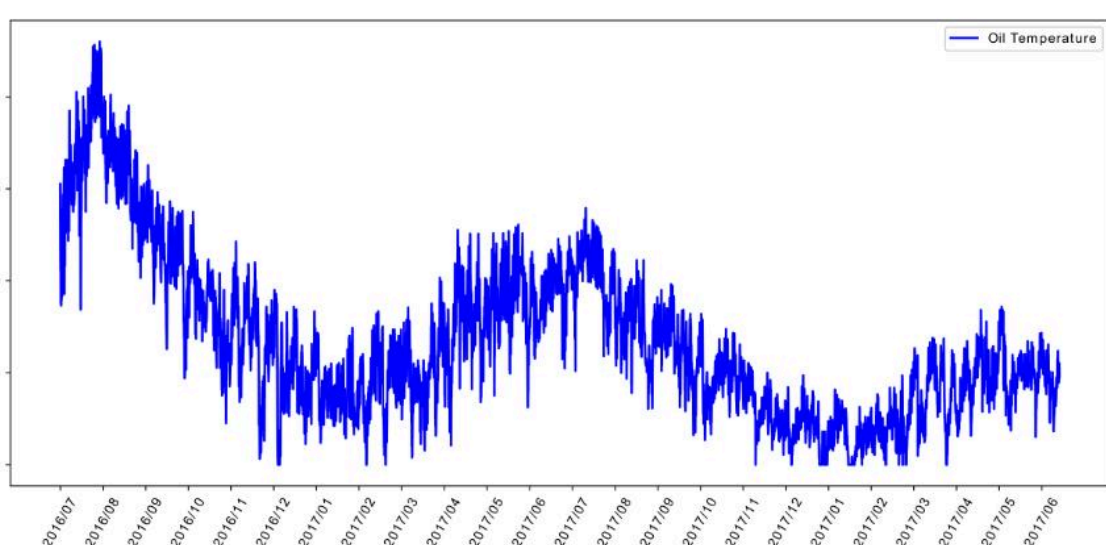


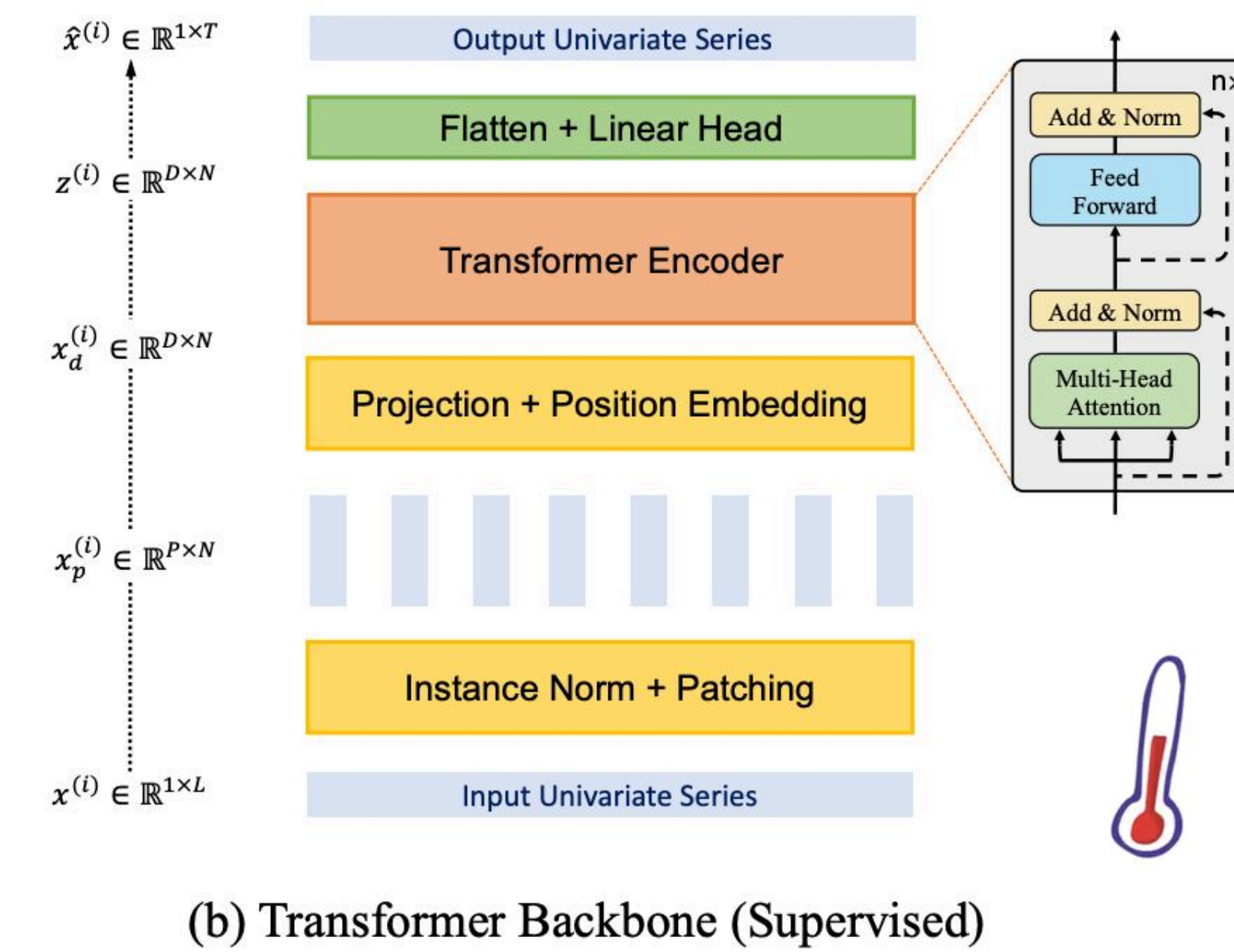
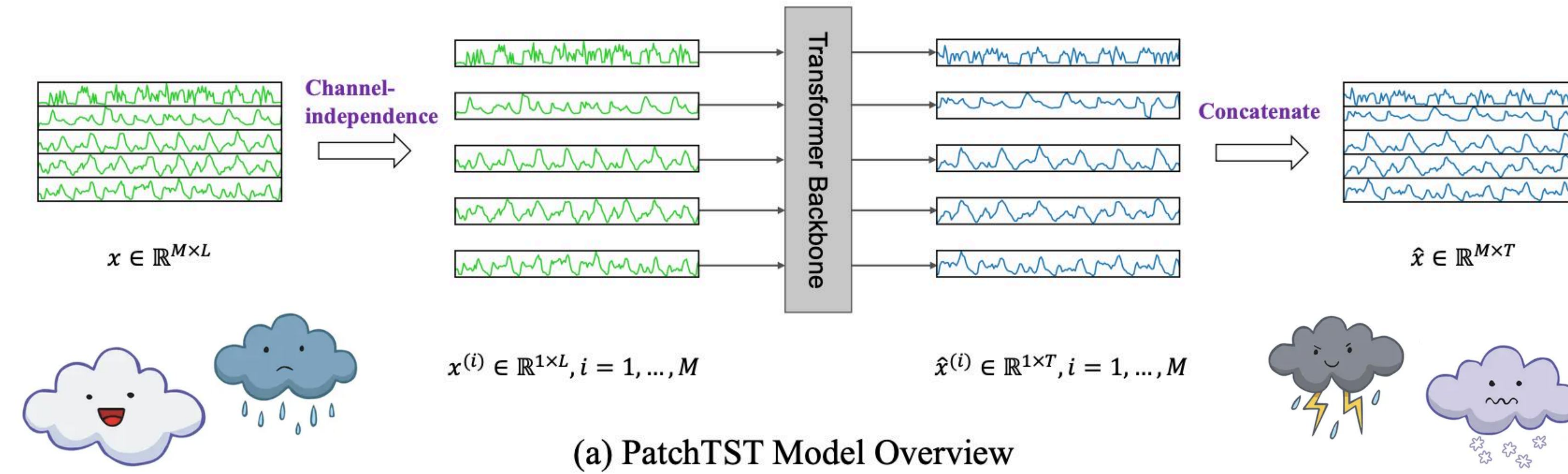
Figure 1. The overall view of "OT" in the ETT-small. Figure 2. The autocorrelation graph of all variables.

A Time Series is Worth 64 Words: Long-term Forecasting with Transformers

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METHODOLOGY



Transformer Encoder

- 3 stacked encoder layers with multi-head self-attention
- Linear prediction head maps flattened patch representations directly to forecast horizon

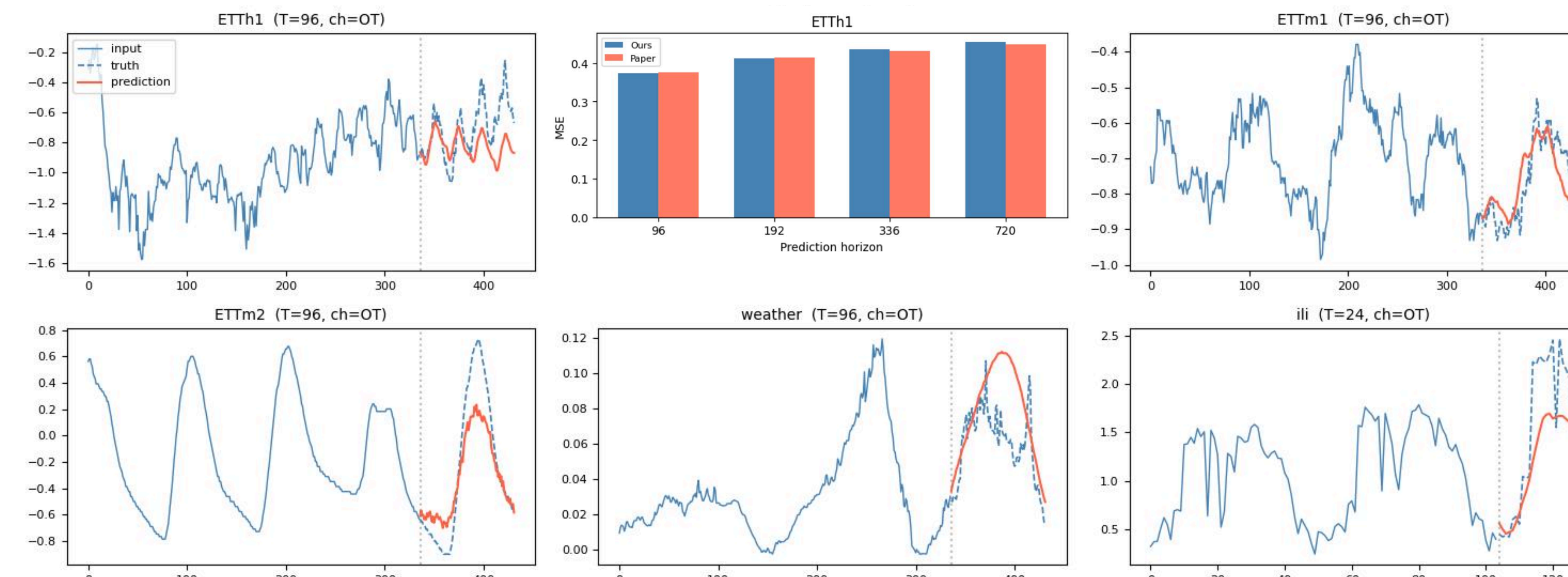
Patching:

- Splits each channel's input into overlapping patches
- Reduces token sequence length, lowering attention cost and increasing local context

Channel Independence:

- Each channel processed separately through shared Transformer weights
- No cross-channel attention; avoids spurious correlations

RESULTS



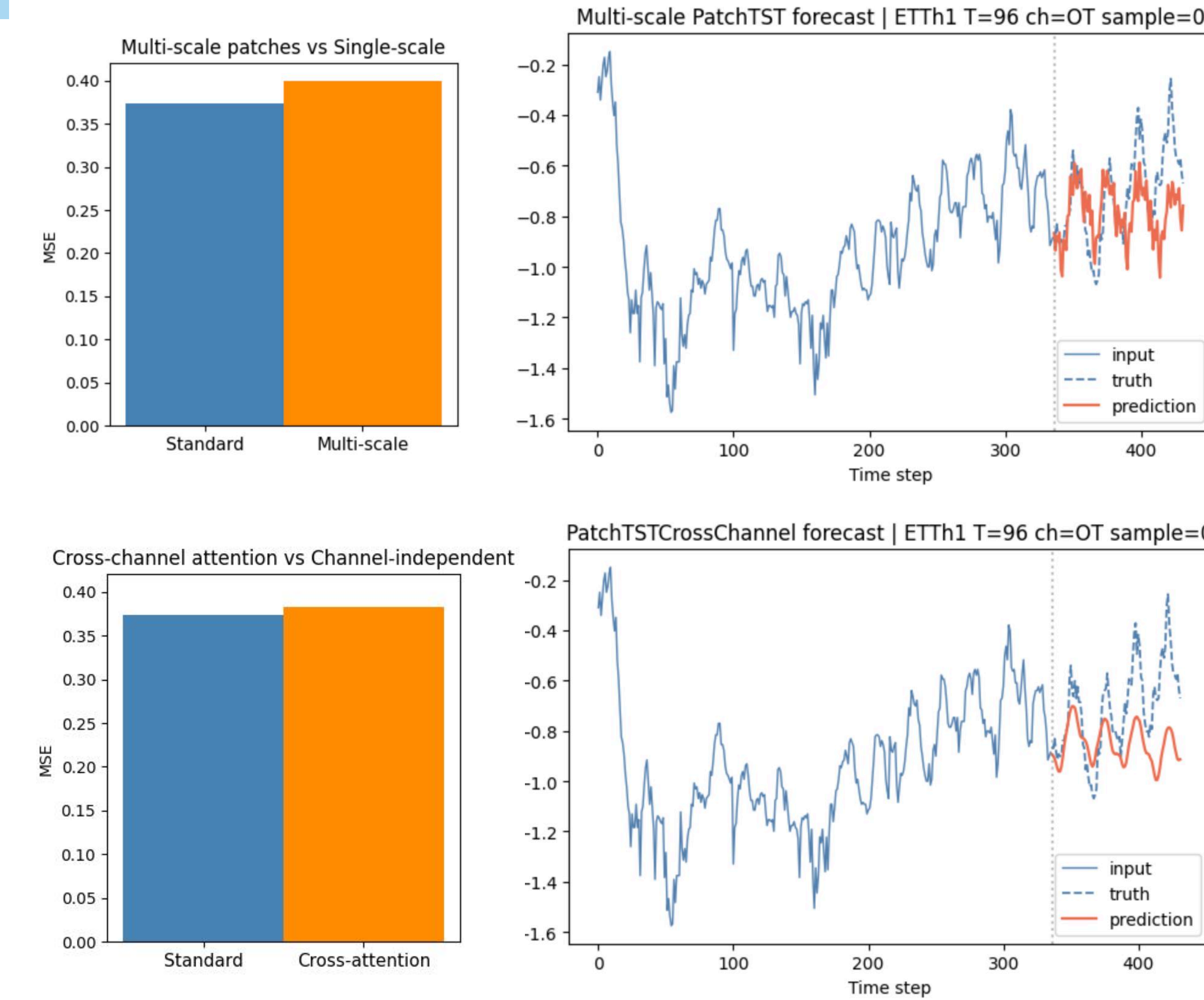
EXTENSIONS

Multi-scale Patching:

- Runs on multiple patch lengths in parallel and fuses at end
- Problems: linear head doesn't capture nuances, redundant

Cross-channel Attention:

- Each token has patch and channel embedding
- Can attend to other channels to predict each channel
- Problems: high attention cost, too much complexity



CONCLUSION

Key Takeaways:

- **Patching** reduces sequence length and improves local temporal context
- **Channel independence** outperforms cross-channel attention by avoiding spurious correlations
- RevIN handles distribution shift between train and test sets

Future Directions:

- Self-supervised pre-training via masked patch modeling (PatchTST/64)

REFERENCES

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). *A time series is worth 64 words: Long-term forecasting with transformers*. International Conference on Learning Representations (ICLR 2023).